# Feature Selection for Depressive Episode Prevalence: Environmental and Socioeconomic Predictors

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## 1 Introduction

In 2026, depression is in the headlines almost daily. From reports of its elevated rate [Wit26], to breakthrough [Mik26] and experimental [US 26] treatments, to newfound triggers [Har26], depression draws sustained attention in modern public discourse. Building on depression's frequent media presence, we note that anyone who has experienced a Wisconsin winter is familiar with the idea of "winter blues" and its clinical cousin, seasonal affective disorder. This naturally leads us to study connections to environmental factors. At the same time, the abundance of remarks about money and happiness motivates us to consider socioeconomic data.

We examine this problem from a public health perspective. Specifically, we let $Y$ denote the prevalence of singular depressive episodes—defined as F32 in ICD-10. Our design matrix $X$ comprises various environmental and socioeconomic features that reflect the considerations above. Our primary goal with feature selection is interpretability rather than reducing overfitting or training time. Certain features within our selected set could inform policy through environmental regulation or socioeconomic legislation.

In section 2 we provide an overview of the SatHealth dataset and describe preprocessing. Then in section 3 we describe our feature selection methods: subsection 3.1 covers correlation screening, subsection 3.2 discusses the LASSO algorithm and MDI, and finally in subsection 3.3 we examine the leave-one-covariate-out (LOCO) procedure. We give the results of these feature selection methods in section 4, and in section 5 we give a final recommendation for the feature set.

All code is available at `https://github.com/spencervenancio/SatHealth_analysis`

## 2 Data and Preprocessing

For our analysis we use SatHealth, a novel dataset combining environmental data, disease prevalences estimated from medical claims, and social determinants of health (SDoH) indicators for Ohio [Wan+25].

### 2.1 The Datasets

The data we use is spread across six datasets: `airquality.csv`, `climate.csv`, `greenery.csv`, and `landcover.csv` give the environmental data at the county level. `sdi.csv` contains the social deprivation index—and additional socioeconomic indicators—for corresponding counties. Finally `icdl3_prev_ohio.csv` gives the prevalences for all ICD10 diseases in Ohio for each Core-based Statistical Areas (CBSAs). We focus on F32, which is a singular depressive episode.

In order to reconcile the differences in geographic level the Census Bureau provides a CBSA to FIPS (county codes) crosswalk. Because CBSA's contain multiple counties we adopt a broadcast: one CBSA observation becomes $N$ county-level rows, each inheriting the same prevalence value.

Another modeling issue is that `airquality`, `climate`, and `greenery` are monthly, while `landcover`, `sdi`, and the target `icdl3_prev_ohio` are annual. In this case we sum additive monthly variables, and take the mean across non-additive variables.

## 2.2 Preprocessing

After the data wrangling described above, we have a dataframe with $n = 333$ observations and $p = 66$ features. Three columns contain either `NA` values or have zero variance, so we drop them entirely. This leaves us with a design matrix with $n = 333$ and $p = 63$. For methods sensitive to feature scale, such as LASSO, we standardize each column to zero mean and unit variance.

# 3 Methods

We evaluate three families of feature selection methods: filter, embedded, and wrapper approaches. For each family we evaluate at least one method.

## 3.1 Filter Methods

Filter methods are the most computationally cheap family. They involve methods which have no connection to prediction functions at all.

### 3.1.1 Correlation Screening

We implement correlation screening using Pearson's $r$. We rank all features by $|\hat{r}_i|$ and retain the top $K = 10$:

$$\widehat{S^*_{\text{corr}}} = \{X_{(1)}, \ldots, X_{(K)}\}$$

where features are ordered by decreasing $|\hat{r}|$.

## 3.2 Embedded Methods

Embedded methods perform feature selection for no additional cost beyond model training. They are, however, limited by the modeling assumptions and limitations associated with that specific model.

### 3.2.1 LASSO

All predictors are standardized to zero mean and unit variance before fitting. Rather than selecting $\lambda$ to minimize prediction error, we target interpretability directly by traversing the full regularization path ($|\Lambda| = 200$ candidate values) and taking the largest $\lambda$ for which at most $K = 10$ features survive:

$$\lambda^* = \max\bigl\{\lambda \in \Lambda : \bigl|\{i : \hat{\beta}_i(\lambda) \neq 0\}\bigr| \leq K\bigr\}.$$

Our feature set is then

$$\widehat{S^*_{\text{lasso}}} = \{X_i : \hat{\beta}_i(\lambda^*) \neq 0\}.$$

Beyond inducing sparsity, the $\ell_1$ penalty resolves the non-identifiability that arises when $X^\top X$ is rank-deficient: among the infinitely many minimizers of the unregularized least-squares objective, LASSO returns the one with smallest $\ell_1$ norm, which in the presence of highly correlated features amounts to selecting a single representative from each correlated cluster.

#### 3.2.2 Mean Decrease in Impurity (MDI)

We also consider an approach using the mean decrease in impurity from a random forests model. The ensemble consisted of $n_{\text{estimators}} = 100$, $\log_2 p$ features were considered at each split, and used bootstrapped sampling. We use the mean decrease in the Gini impurity to return the $K = 10$ most important features.

### 3.3 Wrapper Methods

Wrapper methods are feature selections which utilize information from the prediction function, but are model agnostic. We use a random forest regressor with $n_{\text{estimators}} = 100$, $\log_2 p$ features considered at each split, and bootstrapped sampling. The data are partitioned 80/20 into training and test sets; the baseline model and all leave-one-out refits are trained on the 80% split, and feature importance is the resulting increase in MSE on the held-out 20%.

#### 3.3.1 LOCO

We perform LOCO equipped with loss function $\mathscr{L} := \text{MSE}$. We use it to select the top $K = 10$ features. Under correct model specification, the population importance satisfies $I_j = 0 \iff Y \perp X_j \mid X_{-j}$, so LOCO directly targets the conditional-independence formulation of feature selection ($\tilde{S}$). The flip side, as we will see in section 5, is that highly correlated features are interchangeable substitutes for one another and can drive all of their LOCO importances toward zero.

### 3.4 Stability Analysis

In addition to performing the feature selection above, we would like to estimate how stable these methods are. To do this, we use bootstrapping—resampling with replacement—and apply each method to the perturbed data. We repeat this procedure $B = 50$ times, then calculate the Jaccard index $J(S_1, S_2) = |S_1 \cap S_2|/|S_1 \cup S_2|$ for each of the $\binom{50}{2} = 1225$ pairs of selected sets and average the results.

## 4 Results

We consider each method using the top $K = 10$ features to make them easily comparable. We summarize the selected sets in Table 1, and report stability in Table 2.

## 5 Discussion

The first thing to acknowledge is the correlation between features, as this can explain some of the discrepancies that we see. By analyzing Figure 1 we can observe a classic feature selection phenomenon: within a correlated cluster, correlation screening selects all, LASSO selects one representative, and

| Feature | corr. screening | MDI | LASSO | LOCO |
|---|---|---|---|---|
| County_population | | | | ■ |
| Education_LT12years_score | | | ■ | |
| HHRenter_Occupied_score | | | | ■ |
| SDI_score | | | | ■ |
| bare-coverfraction | | | ■ | |
| dewpoint_temperature_2m | ■ | | | |
| evaporation_from_bare_soil_sum | | | | ■ |
| grass-coverfraction | | ■ | ■ | |
| lake_mix_layer_temperature | ■ | ■ | | |
| leaf_area_index_high_vegetation | | | ■ | |
| leaf_area_index_low_vegetation | ■ | ■ | ■ | ■ |
| ozone | ■ | | | |
| pctHH_Renter_Occupied | | | | ■ |
| shrub-coverfraction | | ■ | | |
| snow_cover | ■ | | | ■ |
| snow_depth | ■ | ■ | ■ | |
| soil_temperature_level_1 | | ■ | | ■ |
| surface_latent_heat_flux_sum | | | | ■ |
| surface_thermal_radiation_downwards_sum | ■ | ■ | ■ | |
| temperature_2m | ■ | | | |
| total_aerosol_optical_depth_at_550nm_surface | | ■ | | |
| urban-coverfraction | | ■ | | |
| volumetric_soil_water_layer_3 | | | ■ | |
| water-permanent-coverfraction | | | | ■ |
| water-seasonal-coverfraction | ■ | | ■ | |
| year | ■ | ■ | ■ | |

Table 1: Feature selection results across methods.

| Method | Jaccard (B=50) |
|---|---|
| corr. screening | 0.542 |
| MDI | 0.357 |
| LASSO | 0.438 |
| LOCO | 0.091 |

Table 2: Jaccard similarity by method.

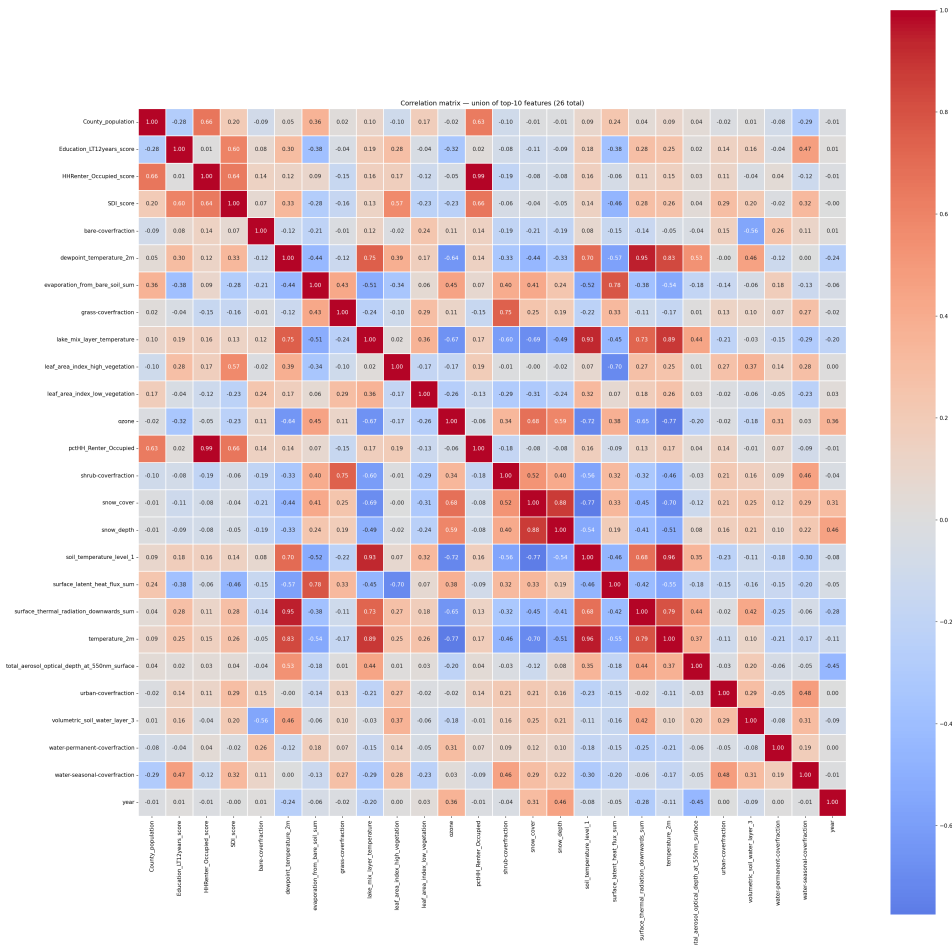


Figure 1: Correlation Matrix of Features Selected by at Least One Method

LOCO selects none. Consider the set of features {`surface_thermal_radiation_downwards_sum`, `dewpoint_temperature_2m`, `lake_mix_layer_temperature`, `temperature_2m`, `ozone`}. LASSO selects just `surface_thermal_radiation_downwards_sum`, correlation screening selects them all, and LOCO selects none. The correlations between `surface_thermal_radiation_downwards_sum` and the others are 0.95, 0.73, 0.68, 0.79, and $-0.65$ respectively.

Because we have so many correlated feature clusters within the data, this is a major drawback for LOCO. LOCO will fail to select any important features within a correlated cluster, since removing one leaves nearly identical predictive information available through its correlated partners. These drawbacks, combined with the very low (0.091) Jaccard index across 50 bootstrapped samples, leave me skeptical of these results.

Another concern is that both correlation screening and LASSO only capture linear relationships. This is a strong assumption given that the dataset spans health, socioeconomic, and environmental domains. These methods do, however, outperform the others in terms of stability, with correlation screening and LASSO having Jaccard indices of 0.542 and 0.438 respectively.

MDI is attractive due to its ability to capture nonlinear relationships, but it has key issues. The dependence between our features causes it to be unstable, which is reflected by the Jaccard index. In the presence of dependent features the initial splits become somewhat random, so under the data perturbations of bootstrapped resampling, MDI produces quite different results. In addition, MDI does not directly measure the conditional independence that defines our feature selection target. Because our main goal was interpretability, the argument that tree models have better predictive accuracy does not give the method a lead over the other methods.

Based on our stated goal of interpretability and imagining that we would like to use these features to create new public policy, I recommend the set produced by *LASSO*. The linearity assumption is its biggest drawback, but this is shared with correlation screening—which only measures marginal correlation. Correlation screening also selects a repetitive set because many of its selections are correlated with each other. LASSO's representative behavior on correlated features aligns with our domain goal. Meanwhile both LOCO and MDI do not handle correlated features well, and their stability scores reflect that. In addition, the coefficients that LASSO provides have intrinsic interpretability, which is a strong advantage.

We considered adding polynomial terms to our LASSO model, but given the shape of our $X$ data frame, even polynomials of degree $d = 2$ produce a new shape of $n = 333$, $p = 2015$, and for $d = 3$ we would face a computationally infeasible $n = 333$, $p = 43679$. Given the tradeoff between flexibility and stability, and given that interpretability is paramount, we accept the linearity assumption as a known limitation rather than pursuing polynomial expansions that would degrade both stability and interpretability.

## 5.1 Limitations

Lastly we would like to note two limitations of our analysis. First, there is a concern over access to care. Our $Y$ is intended as a measure of the prevalence of depression, but the number is derived from medical claims, and wealthier people have much greater access to care and to diagnosis. We observed far fewer socioeconomic features in our selections $\widehat{S^*}$, and this likely plays a factor in that.

Another limitation comes from the broadcasting of the target variable (at the CBSA level) to the existing dataframe (at the FIPS level). This constraint was imposed by the form in which the data were given, but it could have negative effects on our feature selection methods and modeling.

### 5.2 Conclusion

This analysis set out to identify environmental and socioeconomic predictors of F32 prevalence in Ohio, with interpretability for policy as our stated objective. Across four feature selection methods spanning the filter, embedded, and wrapper families, LASSO emerges as the most defensible recommendation. It directly resolves the non-identifiability created by heavily correlated environmental features, returns interpretable coefficients, and demonstrates substantially better stability than either MDI or LOCO. The resulting feature set is dominated by environmental predictors (eg. snow depth, leaf area indices, surface thermal radiation, and several land-cover fractions) while `education_LT12years_score` appears as the only socioeconomic feature. This imbalance is itself informative, but it must be read alongside the access-to-care concern raised in subsection 5.1 because $Y$ is derived from medical claims, it likely under counts depression in populations with weaker access to diagnosis, which would systematically suppress socioeconomic signal regardless of method. Subject to that caveat, our results suggest environmental conditions are at least as detectable as socioeconomic ones in claim-derived F32 prevalence, and any policy work building on this analysis should pair environmental regulation with serious investigation of whether the socioeconomic channel is genuinely weaker or simply hidden by the measurement.

## References


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